

Attention Mechanism: Context Understanding and Limitations

Task 1: Concept Understanding

1. How Attention Helps LLMs Understand Relationships Between Words

The attention mechanism allows an LLM to focus on the most relevant words in a sentence when processing each token. Instead of treating words independently or only looking at nearby words, attention measures how strongly each word relates to the others. This helps the model identify references, dependencies, and the overall meaning of a sentence.

2. Why Attention is Important for Context Understanding

Attention enables the model to capture long-range relationships, understand pronouns, interpret context, and preserve meaning even when important words are far apart. This improves translation, summarization, question answering, and conversational AI.

Task 2: Applied Analysis

1. Why Both Sentences Mean the Same

Sentence A: "The doctor said the patient is stable because he responded well."

Sentence B: "Because he responded well, the doctor said the patient is stable."

Both sentences express the same idea: the patient's positive response is the reason the doctor concluded that the patient is stable. The word order changes, but the meaning remains unchanged.

2. How Attention Should Handle These Sentences

The attention mechanism should connect the words 'responded well' with 'patient' and relate that information to 'stable'. Even though the sentence structure changes, attention should identify the same semantic relationships and produce similar interpretations.

Task 3: Critical Thinking

1. Why the Model Might Still Produce Different Outputs

LLMs learn statistical patterns from training data. Small wording or structural changes may alter attention patterns, causing the model to emphasize different tokens. Ambiguous pronouns, training data bias, or decoding randomness can also lead to different responses.

2. Suggested Improvement

Improve robustness by fine-tuning the model on paraphrase datasets that contain multiple sentence structures with identical meanings. This helps the model learn to produce consistent outputs despite changes in word order.